

MySQL Statement Syntax

To help you find the syntax you need when you need it, this appendix lists the syntax for the most frequently used MySQL operations. For each statement, you'll find a brief description followed by the appropriate syntax. For added convenience, you'll also find cross-references to the chapters where specific statements are taught.

When reading statement syntax, remember the following:

- The | symbol is used to indicate one of several options, so `NULL|NOT NULL` means specify either `NULL` or `NOT NULL`.
- Keywords or clauses enclosed in square brackets [like this] are optional.
- Not all MySQL statements are listed, nor is every clause and option listed.

ALTER TABLE

`ALTER TABLE` is used to update the schema of an existing table. To create a new table, use `CREATE TABLE`. See Chapter 21, “Creating and Manipulating Tables,” for more information.

► Input

```
ALTER TABLE tablename
(
    ADD column datatype [NULL|NOT NULL] [CONSTRAINTS],
    CHANGE column columns datatype [NULL|NOT NULL] [CONSTRAINTS],
    DROP column,
    ...
);
```

COMMIT

`COMMIT` is used to write a transaction to a database. See Chapter 26, “Managing Transaction Processing,” for more information.

► Input

```
| COMMIT;
```

CREATE INDEX

CREATE INDEX is used to create an index on one or more columns. See Chapter 21 for more information.

► Input

```
CREATE INDEX indexname  
ON tablename (column [ASC|DESC], ...);
```

CREATE PROCEDURE

CREATE PROCEDURE is used to create a stored procedure. See Chapter 23, “Working with Stored Procedures,” for more information.

► Input

```
CREATE PROCEDURE procedurename( [parameters] )  
BEGIN  
...  
END;
```

CREATE TABLE

CREATE TABLE is used to create a new database table. (To update the schema of an existing table, use ALTER TABLE.) See Chapter 21 for more information.

► Input

```
CREATE TABLE tablename  
(  
    Column datatype [NULL|NOT NULL] [CONSTRAINTS],  
    Column datatype [NULL|NOT NULL] [CONSTRAINTS],  
    ...  
);
```

CREATE USER

CREATE USER is used to add a new user account to the system. See Chapter 28, “Managing Security,” for more information.

► Input

```
CREATE USER username[@hostname]  
[IDENTIFIED BY [PASSWORD] 'password'];
```

CREATE VIEW

CREATE VIEW is used to create a new view of one or more tables. See Chapter 22, “Using Views,” for more information.

► Input

```
CREATE [OR REPLACE] VIEW viewname  
AS  
SELECT ...;
```

DELETE

DELETE deletes one or more rows from a table. See Chapter 20, “Updating and Deleting Data,” for more information.

► Input

```
DELETE FROM tablename  
[WHERE ...];
```

DROP

DROP permanently removes database objects (tables, views, indexes, and so forth). See Chapters 21, 22, 23, 24, “Using Cursors,” 26, and 28 for more information.

► Input

```
DROP DATABASE|INDEX|PROCEDURE|TABLE|TRIGGER|USER|VIEW  
itemname;
```

INSERT

INSERT adds a single row to a table. See Chapter 19, “Inserting Data,” for more information.

► Input

```
INSERT INTO tablename [(columns, ...)]  
VALUES(values, ...);
```

INSERT SELECT

INSERT SELECT inserts the results of a SELECT into a table. See Chapter 19 for more information.

► Input

```
INSERT INTO tablename [(columns, ...)]  
SELECT columns, ... FROM tablename, ...  
[WHERE ...];
```

ROLLBACK

ROLLBACK is used to undo a transaction block. See Chapter 26 for more information.

► Input

```
| ROLLBACK [ TO savepointname];
```

SAVEPOINT

SAVEPOINT defines a savepoint for use with a ROLLBACK statement. See Chapter 26 for more information.

► Input

```
| SAVEPOINT sp1
```

SELECT

SELECT is used to retrieve data from one or more tables (or views). See Chapter 4, “Retrieving Data,” Chapter 5, “Sorting Retrieved Data,” and Chapter 6, “Filtering Data,” for more basic information. (Chapters 4–17 also cover aspects of SELECT.)

► Input

```
| SELECT columnname, ...  
| FROM tablename, ...  
| [WHERE ...]  
| [UNION ...]  
| [GROUP BY ...]  
| [HAVING ...]  
| [ORDER BY ...];  
| ;
```

START TRANSACTION

START TRANSACTION is used to start a new transaction block. See Chapter 26 for more information.

► Input

```
| START TRANSACTION;
```

UPDATE

UPDATE updates one or more rows in a table. See Chapter 20 for more information.

► Input

```
| UPDATE tablename  
| SET columnname = value, ...  
| [WHERE ...];
```